DIPOLE TYPE BY LENGTH

STEP 0 — always start here *λ = wavelength (m), l = antenna rod length (m), $c=3\times10^8$ m/s*

$\boxed{\lambda = \frac{c}{f}\ \text{(m)}}$ ratio $\frac{l}{\lambda}$ → pick row below

l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R,\theta)$, $D=1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave shortcut, $D=1.64$, $R_{\text{rad}}=73\,\Omega$
$= 1$	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}}=199\,\Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

HERTZIAN DIPOLE — $S(R,\theta)$

Far-field E,H phasors *small $l \ll \lambda$*

$$\tilde{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi}\frac{e^{-jkR}}{R}\sin\theta, \quad \tilde{H}_\phi = \tilde{E}_\theta/\eta_0$$

Power density *$l/\lambda < 1/50$; far field*

$$S(R,\theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}\sin^2\theta = S_{\max}\sin^2\theta\ (\text{W/m}^2)$$

l rod length (m); I_0 peak current (A); R obs. dist. (m); θ from dipole axis; $k=2\pi/\lambda$ (rad/m); $\eta_0=120\pi\approx377\,\Omega$.

Max @ broadside $\theta_{\max}=\pi/2$; D *exact* $D=1.5$ (1.76 dB)

Total radiated power $P_{\text{rad}} = \frac{\eta_0\pi}{3}(I_0 l/\lambda)^2$ (W)

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R *any l*

$$S(\theta) = \frac{15I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right)-\cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2$$

At $\theta=0,\pi$: bracket is 0/0. L'Hôpital ⇒ $S=0$ (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta=0.001$ rad → tiny → 0.

Normalized pattern *dimensionless* $F(\theta)=S(\theta)/S_{\max}$

Quarter-wave $l=\lambda/4$ *$\pi l/\lambda=\pi/4$, $\cos(\pi/4)=\sqrt{2}/2$*

$$\frac{S(\theta)}{S_0} = \left[\frac{\cos(\frac{\pi}{4}\cos\theta)-\frac{\sqrt{2}}{2}}{\sin\theta}\right]^2, \quad S_0=\frac{15I_0^2}{\pi R^2}$$

$$S_{\max} = S(\pi/2) = S_0\left(1-\frac{\sqrt{2}}{2}\right)^2 \approx 0.0858\,S_0$$

$D=1.64$, $R_{\text{rad}}=36.5\,\Omega$, $\theta_{\max}=\pi/2$.

Half-wave $l=\lambda/2$ *$\pi l/\lambda=\pi/2$, shortcut form*

$$S(\theta) = \frac{15I_0^2}{\pi R^2}\frac{\cos^2(\frac{\pi}{2}\cos\theta)}{\sin^2\theta}$$

$D=1.64$, $R_{\text{rad}}=73\,\Omega$, $\theta_{\max}=\pi/2$.

COMMON ANGLES (deg ↔ rad, trig values)

Conversion *multiply by $\pi/180$ or $180/\pi$*

$$\theta_{\text{rad}} = \theta_{\text{deg}}\cdot\frac{\pi}{180} \qquad \theta_{\text{deg}} = \theta_{\text{rad}}\cdot\frac{180}{\pi}$$

°	rad	$\sin\theta$	$\cos\theta$	$\sin^2\theta$	$\cos^2\theta$
0	0	0	1	0	1
30	$\pi/6$	$1/2$	$\sqrt{3}/2$	$1/4$	$3/4$
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	$1/2$	$1/2$
60	$\pi/3$	$\sqrt{3}/2$	$1/2$	$3/4$	$1/4$
90	$\pi/2$	1	0	1	0
180	π	0	-1	0	1

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

Setup *N identical elements along axis, spacing d*

$$\text{elem pos } z_i = id,\ i=0,\dots,N-1;\ A_i = a_i e^{j\psi_i}$$

Equal phase ($\psi_i=0$) ⇒ broadside beam ($\theta_{\max}=\pi/2$). Progressive phase $\psi_i = i\alpha$ ⇒ steered beam at $\cos\theta_{\max} = -\alpha/(kd)$.

Pattern multiplication *S_e = single element pattern*

$$S(R,\theta,\phi) = S_e(R,\theta,\phi)\,F_a(\theta)$$

Array factor (general) *$k=2\pi/\lambda$*

$$F_a(\theta) = \left|\sum_{i=0}^{N-1} A_i\,e^{jkid\cos\theta}\right|^2$$

Uniform $A_i=1$, equal phase *closed form*

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)},\ \gamma = kd\cos\theta$$

Peak: $F_a^{\max}=N^2$ at $\gamma=0$ (broadside, $\theta=\pi/2$). Normalize: $F_a^{norm}=F_a/N^2$.

HPBW (broadside, uniform) *use Nd, not $(N-1)d$*

$$\beta \approx 0.88\,\frac{\lambda}{Nd}\ (\text{rad}) = 50.8^\circ\cdot\frac{\lambda}{Nd}$$

Grating lobes appear if $d>\lambda$ (broadside). Avoid.

BEAM PARAMETERS — β, Ω_p, D

Normalize *always start here*

$$F(\theta) = S(\theta)/S_{\max}\ \text{(unitless)}$$

Beamwidth β at threshold X

Set $F(\theta_X)=X$, solve for θ_X; $\beta=2\theta_X$ (symmetric)

Threshold	$X=10^{\text{dB}/10}$	θ_X (Gauss. short-cut)
HPBW (-3 dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
-6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
-10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
any $F(\theta_X)$	any X	$F(\theta_X)=X$

Pattern solid angle Ω_p *recognize pattern, look up*

$$\Omega_p = \int_0^{2\pi}\int_0^{\pi}F(\theta)\sin\theta\,d\theta\,d\phi\ \ (\text{sr})$$

No ϕ dep.: $\Omega_p = 2\pi\int_0^\pi F(\theta)\sin\theta\,d\theta$.

θ is the integration variable — DON'T plug in a number for θ (e.g. $\theta_{1/2}$). Integrate over $\theta\in[0,\pi]$.

Pattern $F(\theta)$	Ω_p (sr)
Isotropic ($F=1$)	$4\pi\approx12.57$
Hertzian $\sin^2\theta$	$8\pi/3\approx8.38$
Half-wave dipole	≈ 7.66
Narrow Gauss $e^{-a\theta^2}$	$\pi/a=\pi\theta_{1/2}^2/\ln 2$
2-axis HPBW (aperture)	$\beta_{xz}\cdot\beta_{yz}$
Other (use integral)	numerical

TI-36X Pro: `[2nd][d/dx]` for $\int_\square^\square$, RAD mode, multiply by 2π.

Directivity D *always $D=4\pi/\Omega_p$; common shortcuts below*

$$D = \frac{4\pi}{\Omega_p}\ \text{(unitless)} \qquad D_{\text{dB}} = 10\log_{10} D$$

WORD TRAP: "direction of max radiation" = angle θ_{max} (rad/deg). "directivity" = unitless number D. Different things!

Dipoles → COMMON DIPOLE PARAMS table. Aperture: $D\approx4\pi A_p/\lambda^2$. 2-axis HPBW: $D\approx4\pi/(\beta_{xz}\beta_{yz})$. Circular: $D\approx4\pi/\beta^2$.

Gain (if asked) *ξ = rad. eff., $0<\xi\le1$*

$$G=\xi D \qquad G_{\text{dB}}=10\log_{10}G$$

Lossless / ideal antenna ⇒ $G=D$.

COMMON DIPOLE PARAMETERS

Type	l	D	D_{dB}	R_{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$<\lambda/50$	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
Monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{rad}=36.6I_0^2$ W. Monopole ($\lambda/4$ above ground): same patt. as half-wave but P_{rad}, R_{rad} halved.

Lossless antenna ($\xi=1$) ⇒ $G=D$. So in Friis, use G_t, G_r = these D values (e.g. half-wave $G=1.64$).

EFFECTIVE AREA A_e

Definition *antenna's RX cross section (m²); D from BEAM PARAMS or table above*

$$A_e = \frac{\lambda^2 D}{4\pi}\ (\text{m}^2)$$

Physical cross section *wire dipole, diameter d; l see table above*

$$A_p = l\,d\ \ (\text{m}^2\text{, any dipole})$$

Inverse: D from A_e

$$D = \frac{4\pi A_e}{\lambda^2}\ \text{(unitless)}$$

Hertzian special case *$D=1.5$*

$$A_e^{\text{Hertz}} = \frac{3\lambda^2}{8\pi}\approx0.119\lambda^2\ (\text{m}^2)$$

Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim\lambda$.

FRIIS TRANSMISSION FORMULA

Pwr density at RX, then RX pwr *G_t, G_r linear gains; $\lambda=c/f$*

$$S_r = \frac{G_t P_t}{4\pi R^2}\ (\text{W/m}^2);\ \boxed{P_r = G_t G_r\left(\frac{\lambda}{4\pi R}\right)^2 P_t}\ (\text{W})$$

Equivalent: $P_r = S_r A_{e,r}$.

Solve for R *max range*

$$R = \frac{\lambda}{4\pi}\sqrt{\frac{G_t G_r P_t}{P_r}}\ (\text{m})$$

Equivalent form using A_e *recv. area form*

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2},\quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t/L_{fs}$.

General form (off-axis) *when patterns NOT peak-aligned*

$$P_r = G_t G_r\left(\frac{\lambda}{4\pi R}\right)^2 P_t\,F_t(\theta_t,\phi_t)F_r(\theta_r,\phi_r)$$

F_t, F_r normalized patterns at the link directions; $F=1$ when aligned.

Recipe 1. $\lambda=c/f$. 2. Convert dB gains to linear: $G=10^{G_{\text{dB}}/10}$. 3. If TX is "half-wave dipole" use $G_t=1.64$. 4. Plug into Friis.

Use linear G, not dB. Convert FIRST.

dB ↔ LINEAR

Works for any power-like quantity: G gain, D directivity, P_r/P_t, SNR, etc.

$$X_{\text{dB}} = 10\log_{10}X_{\text{lin}} \qquad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

e.g. $D_{dB}=10\log_{10}D$; $G_{dB}=10\log_{10}G$; $SNR_{dB}=10\log_{10}(P_r/P_N)$.

dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
-3	0.5	-10	0.1

Memorize: 3 dB ⇔ ×2, 10 dB ⇔ ×10. Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (power ∝ E^2).

NOISE & SNR

Noise power *thermal noise into matched receiver*

$$P_N = k_B\,T_{\text{sys}}\,B\ (\text{W})$$

$k_B=1.38\times10^{-23}$ J/K; T_{sys} system noise temp (K); B receiver bandwidth (Hz).

Signal-to-noise ratio

$$\text{SNR} = \frac{P_{\text{rec}}}{P_N}\ \text{(unitless)} \quad \text{SNR}_{\text{dB}} = 10\log_{10}\frac{P_{\text{rec}}}{P_N}$$

Typical comm. link needs SNR > 10 dB for usable signal.

APERTURE ANTENNAS (horn, dish)

A_p physical aperture area (m²); A_e effective area (m², $=\lambda^2 D/4\pi$); β_{xz},β_{yz} HPBWs in two planes (rad); ℓ, d aperture side / diameter (m).

Directivity (any shape, uniform illum.)

$$D\approx\frac{4\pi A_p}{\lambda^2}\ \text{(unitless)} \quad \Rightarrow\ A_e\approx A_p$$

Directivity from beamwidths *any aperture*

$$D\approx\frac{4\pi}{\beta_{xz}\beta_{yz}}\ \text{(use rad); circular: } D\approx4\pi/\beta^2$$

HPBW and D by aperture shape *uniform illum.; β in rad*

Shape	A_p	HPBW (rad)	D (unitless)
Rect. $\ell_x\times\ell_y$	$\ell_x\ell_y$	$\beta_x\approx0.88\lambda/\ell_x$, $\beta_y\approx0.88\lambda/\ell_y$	$D\approx4\pi/(\beta_x\beta_y)$ $=4\pi\ell_x\ell_y/\lambda^2$
Square ℓ	ℓ^2	$\beta\approx0.88\lambda/\ell$	$D\approx4\pi/\beta^2 = 4\pi\ell^2/\lambda^2$
Circular (diam. d)	$\pi d^2/4$	$\beta\approx1.02\lambda/d$	$D\approx4\pi/\beta^2 = \pi^2 d^2/\lambda^2$

Scaling rules *$D\propto A_p/\lambda^2$; $\beta\propto\lambda/\sqrt{A_p}$*

Change	New D	New β
Double area ($A_p\to2A_p$)	$D_{\text{new}}=2D_{\text{old}}$	$\beta_{\text{new}}=\beta_{\text{old}}/\sqrt{2}$
Double freq ($\lambda\to\lambda/2$)	$D_{\text{new}}=4D_{\text{old}}$	$\beta_{\text{new}}=\beta_{\text{old}}/2$

DIPOLE

l — antenna length (m)
$\lambda=c/f$ (m); $c=3\times10^8$ m/s
I_0 — peak current (A)
$k=2\pi/\lambda$ (rad/m)
R — obs. dist. (m); θ from axis
$\eta_0=120\pi\approx377\,\Omega$
$S(R,\theta)$ — pwr density (W/m²)

BEAM

$F(\theta)=S/S_{\max}$ (unitless)
a — coef. on θ^2 in Gaussian
β — HPBW (rad or °)
Ω_p — pattern solid angle (sr)
$D=4\pi/\Omega_p$ — directivity
ξ — rad. efficiency; $G=\xi D$
$D_{\text{dB}}=10\log_{10}D$

FRIIS / dB

P_t, P_r — TX/RX power (W)
G_t, G_r — linear gains
$S_r=G_tP_t/(4\pi R^2)$ (W/m²)
$P_r=G_tG_r(\lambda/4\pi R)^2P_t$
$G=10^{G_{\text{dB}}/10}$
3 dB $=\times2$, 10 dB $=\times10$
$P_N=k_BT_{\text{sys}}B$ (W)

APERTURE

A_p — physical area (m²)
$A_e=\lambda^2D/(4\pi)$ (m²)
$A_e\approx A_p$ for aperture
$D\approx4\pi A_p/\lambda^2$ (unitless)
$\beta\approx0.88\lambda/\ell$ (rect/sq.)
$\beta\approx1.02\lambda/d$ (circular)
$2A_p\Rightarrow2D$, $\beta/\sqrt{2}$

ARRAY

N — number of elements
d — spacing (m)
$\gamma=kd\cos\theta$
$F_a=\sin^2(N\gamma/2)/\sin^2(\gamma/2)$
$F_a^{\max}=N^2$ at $\theta=\pi/2$
$F_a^{\text{norm}}=F_a/N^2$
$\beta\approx0.88\lambda/(Nd)$ (broadside)